Tony's SoXR Unity Toolkit Docs (v 1.0)

This collection of Unity C# scripts allows for intuitive actions to take place within a Unity Scene. Most following the format of *if this then that*, along with timing features, collision and "*trigger*" detection.

We use this toolkit like any other script in Unity: first, you find a `game object` to attach the script to, then find `Add Component` and search the name of the script you'd like to use (assuming it's in the project files already).
It's important to note that the script doesn't necessarily have to effect the `game object` it's attached to. For example, you can added a `Timed Action` script to a `game object`, the action that gets preformed after the time (more explanation below) does not have be related to the object the script is attached to. You can call an action on any `game object` in the scene. This means one could have a empty `game object` solely for organizing Unity scripts into one coherent spot.

So! Let's look at our scripts and what they do.

XR Grab Interactable

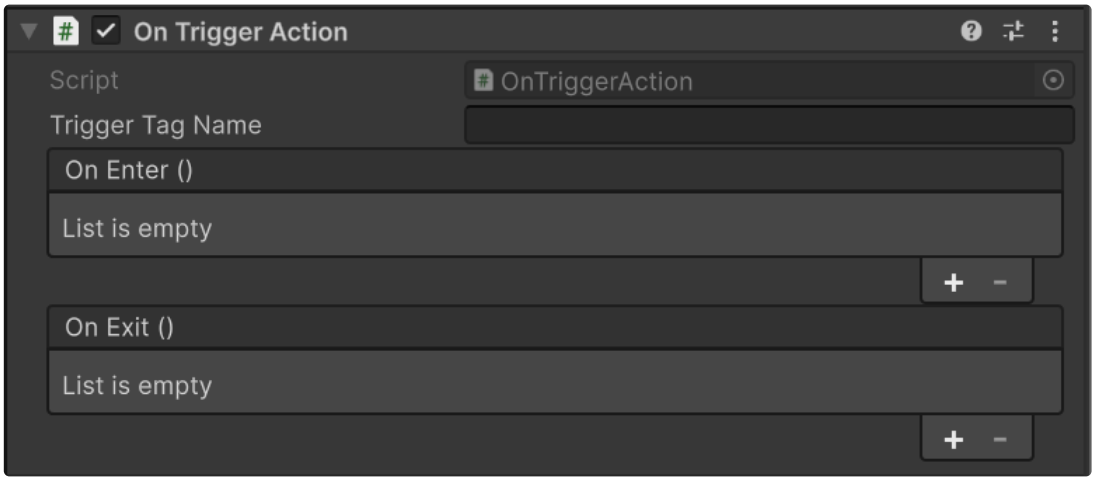
This enables the user to grab an object in XR and move it around. If the object has gravity enabled in the `Rigidbody` component, the object can be thrown by the user by default.

This script is included with the XR package in Unity by default and has a lot of options, read more about the available settings here.

OnTriggerAction

This script detects when `collider` that has the `isTrigger` setting enabled has entered the `collider` of the component's `game object`, then plays an `action`.

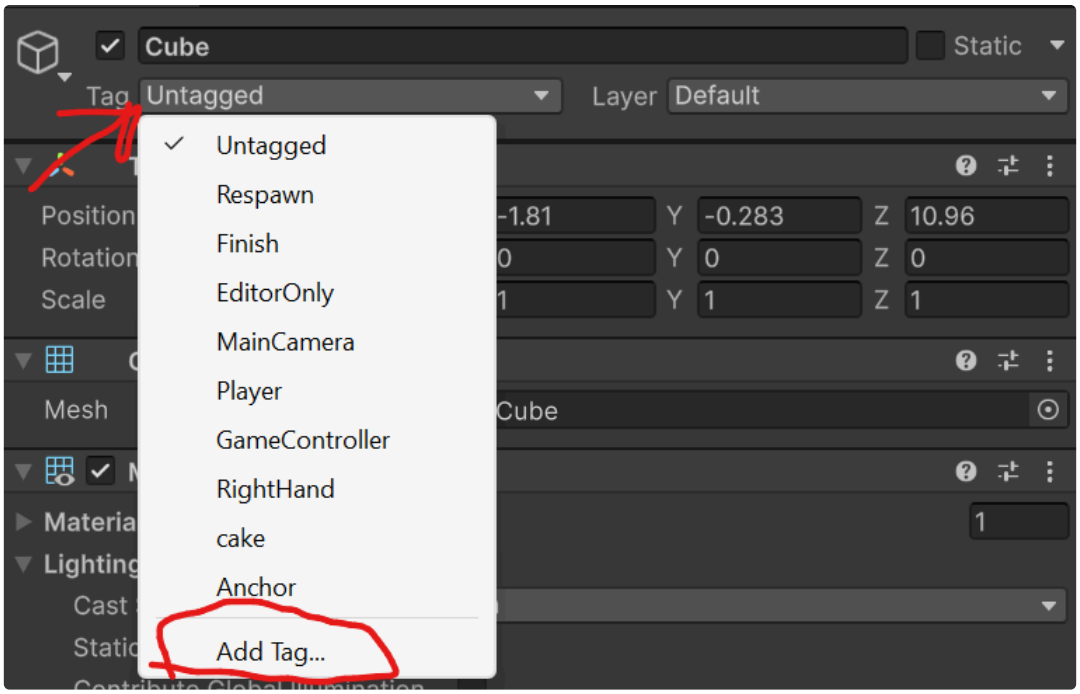
Use this when you want to detect when something enters a particular zone, like a player, and you want something to happen as a result. Like switching music when the player enters a new room.



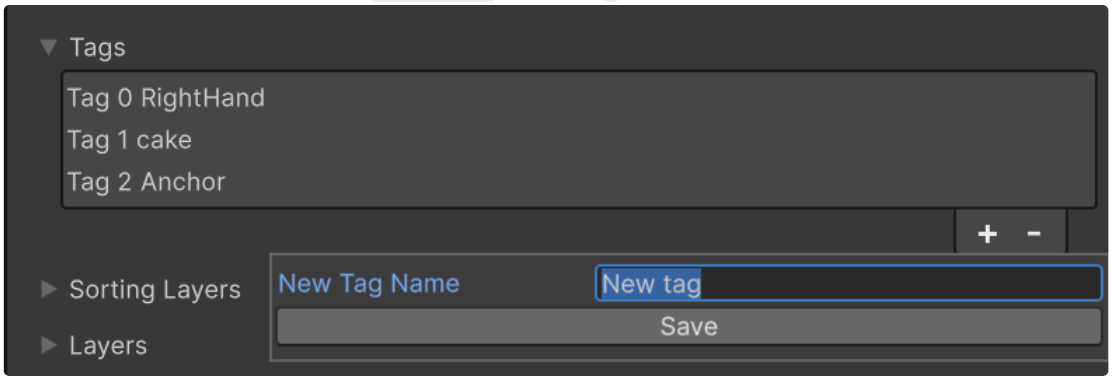
Here's what you'll see once you add the script to a particular `game object`...

Trigger Tag Name

The `Trigger Tag Name` is the name of the tag you want to detect if it's entered the collision zone. You can set any `game object`'s tag to something else in the inspector, as seen below.



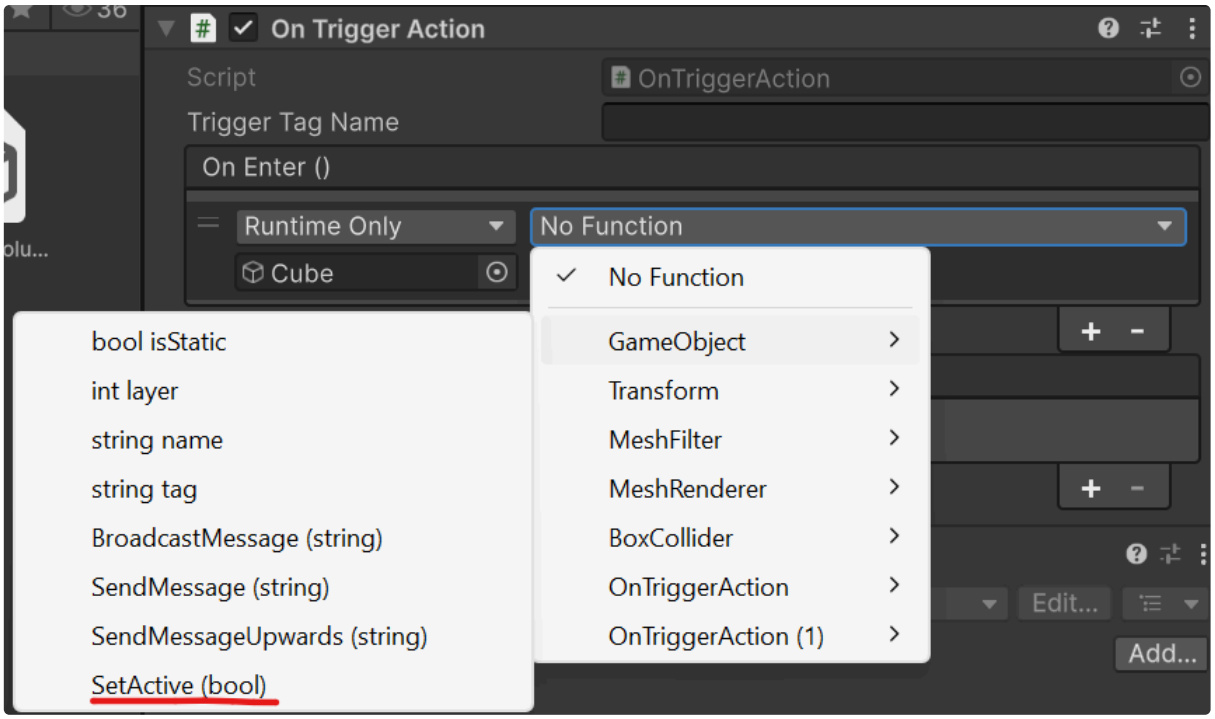
If you want to add a new tag, click `Add Tag...` , find the `+` sign, name your tag, then hit save!



On Enter / On Exit

These are the actions to preform when the `game object` with your specified tag enters the collider field. `On Enter` preforms the actions upon entering the field, and `On Exit` intuitively preforms the action when the object is leaving. You can add any amount of actions to preform.

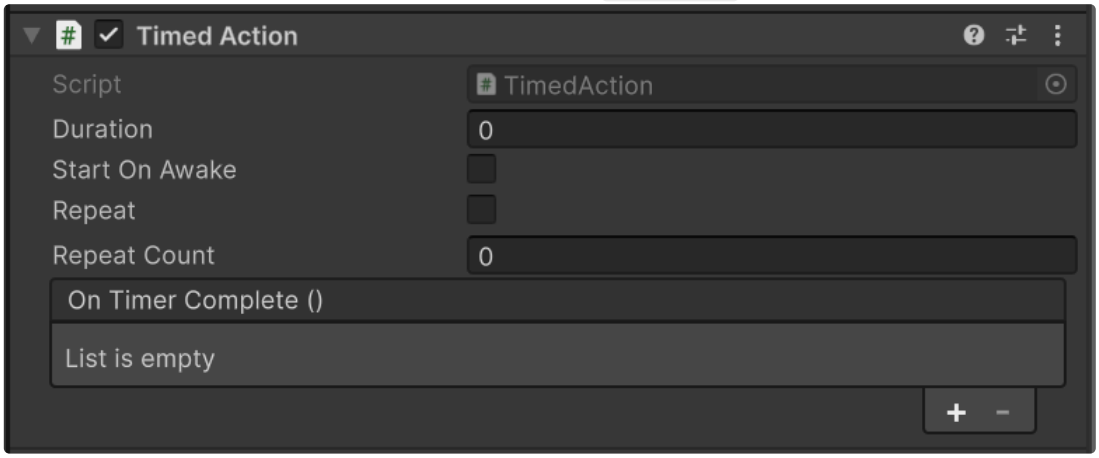
To add an action, drag a `game object` into the object field by dragging it on or clicking the circle. Then, select an action you want to preform; as an example this action will turn on or off the object in the scene.



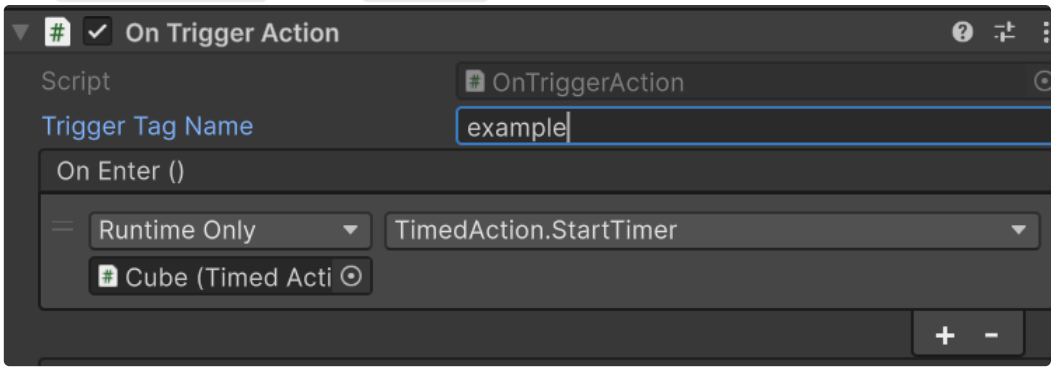
Timed Action

This script will preform an action after a specified amount of seconds for a specified amount of times. It also has the option of infinitely repeating an action.
One could use this script to implement a hint after a certain time period has passed in order to guide the player. Perhaps it could play an audio track to guide the player towards a direction or aid immersion.

Here's what you'll see after you add the script to a given `game object`:



To call this script, you have another `game object` call `StartTimer` to begin the countdown till the action. Here's an example setup with `On Trigger Action` running `StartTimer`.



Duration

This is the amount of time before the action plays in seconds.

Start on Awake

If this is checked, the countdown toward the action start *immediately* instead of needing to be called by another action.

Repeat

If this is checked, the action will repeat itself with the same duration between actions.

Repeat Count

This is the amount of times the action will be preformed if repeat is enabled. If this is set to `0`, the action repeats *infintely*.

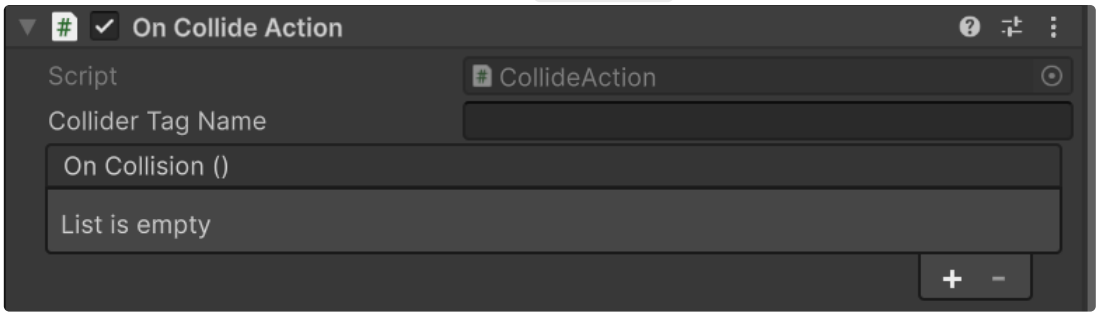
On Timer Complete

The list of actions to preform after the duration.

Collide Action

This script detects detects collision between two `colliders`.
One could use this to detect when a key enters the area of a door key, or a ball hits a target area on a wall.

Here's what you'll see when adding the script to a `game object`:



Collider Tag Name

The name of the other `collider`'s `game object` tag to look out for. This script should be added to the object of which would be the "subject" of the collision.

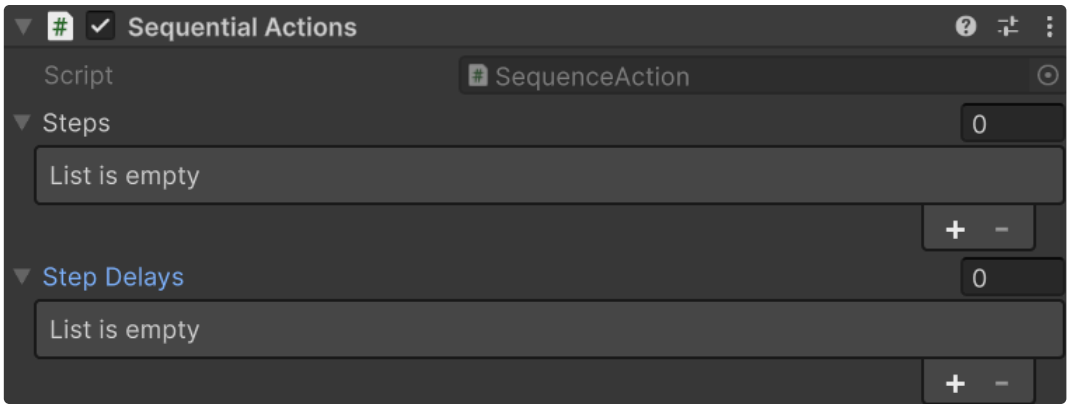
On Collision

This is the list of actions to preform when a collision is detected

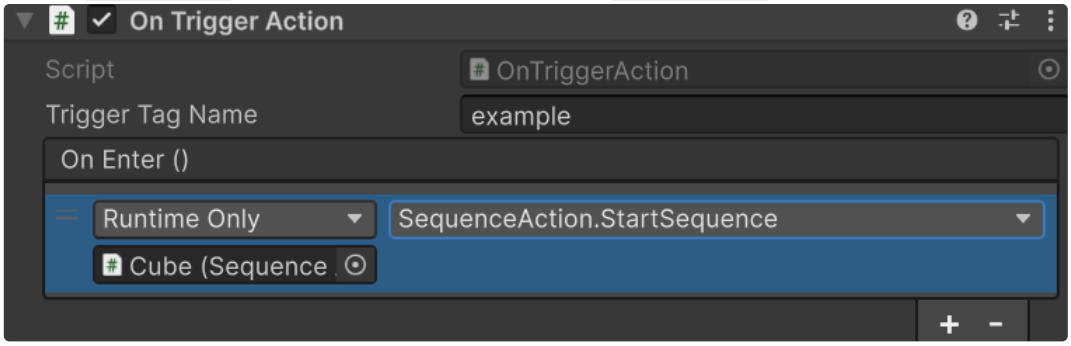
Sequence Action

This script enables you to preform a list of actions one after another, all with their own delays in-between. For example, you could turn on a green light after 4 seconds, then turn on a yellow light after 10 seconds, then a red light after 3 seconds.

This is what you'll see after adding the script to a `game object`:



To run this script, call `StartSequence` from another script. Calling `StartSequence` again after starting will reset the sequence from the beginning.
Calling `StopSequence` will pause the actions, then calling `StartSequence` again will resume the actions being played.



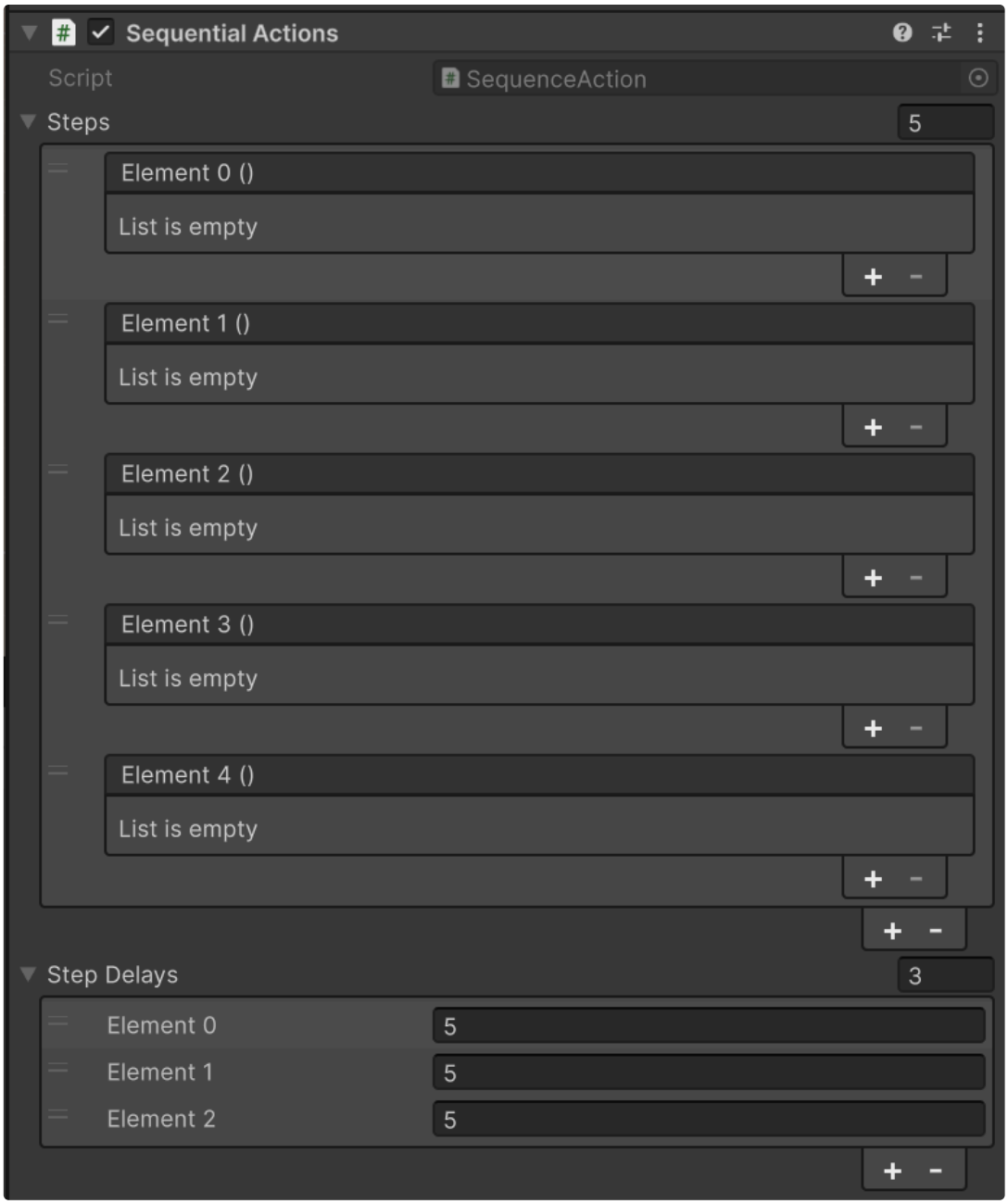
Steps

These are the actions you want to play in order from first to last.

Step Delays

These are the delays you want before each step (or action). If there isn't a delay that's set to a particular step, then the delay is assumed to be zero.

For example, this setup will play 5 actions. The first 3 will have 5 seconds in-between them, and the last 2 actions will play immediately after the third action.


Sequential Actions
Script
SequenceAction
Steps
5
Element 0 ()
List is empty
Element 1 ()
List is empty
Element 2 ()
List is empty
Element 3 ()
List is empty
Element 4 ()
List is empty
Step Delays
3
Element 0
5
Element 1
5
Element 2
5